Naman Goyal

Final-year B.S. (Research) student in Physics, Indian Institute of Science, Bengaluru

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Research Interests

Theoretical high-energy and mathematical physics, focused on the fundamental constraints imposed by quantum field theory and symmetry principles. I combine analytical methods with scientific computing to study **electroweak vacuum stability** and **semiclassical vacuum decay** (three-loop Standard Model running, bounce actions), **effective descriptions of gravitation** (pseudo-Newtonian tests of general relativity, published in *Eur. Phys. J. C*) and **scattering amplitudes** (currently, S-matrix bootstrap constraints on dual-resonance amplitudes), aiming to understand the mathematical structure of fundamental interactions.

Education

Indian Institute of Science (IISc), Bengaluru — **Aug 2023 – Apr 2027 (expected)**
B.S. (Research) in Physics | Cumulative GPA **9.4 / 10**

Modern Vidya Niketan (MVN), Faridabad, Haryana — **Apr 2020 – Apr 2023**
Higher Secondary Schooling (CBSE) | Class X: 97.8% | Class XII: 94.4%

Publications

Tests of general relativity in pseudo-Newtonian approach — **Peer-reviewed | Nov 2026**
N. Goyal, B. Mukhopadhyay and A. K. Meena, Eur. Phys. J. C **86**, 1280 (2026)
doi:10.1140/epjc/s10052-026-16547-6 | arXiv:2602.20261 [gr-qc]

Research Experience

Scattering Amplitudes and the S-Matrix Bootstrap — May 2026 – present

Supervisor: Prof. Rick Sandeepan Gupta, Tata Institute of Fundamental Research (TIFR), Mumbai

- *Question.* Investigating whether the **uniqueness of dual-resonance scattering amplitudes** can be established directly from analyticity, crossing symmetry, unitarity and Regge behavior, avoiding auxiliary assumptions on their zero distribution.
- *Methods.* Studying the analytic properties of scattering amplitudes: polynomial boundedness, **positivity constraints** and the analytic structure in the **complexified Mandelstam plane**, drawing on the modern S-matrix bootstrap program.

Higgs Vacuum Stability and the Cosmological Hierarchy Problem — May 2024 – May 2026

Supervisor: Prof. Rick Sandeepan Gupta, TIFR, Mumbai | Code: github.com/namangoyal31/SMVac

- *Framework.* Developed **SMVac**, a modular C++ framework for precision studies of electroweak vacuum stability: **NNLO electroweak matching** conditions; **three-loop Standard Model β-functions** for the gauge couplings (g_1, g_2, g_3), the top Yukawa y_t and the Higgs quartic λ, RK4-evolved to the **Planck scale**; the one-loop **Coleman-Weinberg effective potential**; and semiclassical vacuum-decay calculations.
- *Decay rates.* Computed Higgs vacuum decay rates from both the analytical **Fubini-Lipatov formalism** and a numerical **RG-improved treatment** that integrates the full running one-loop effective potential over the conformal bounce profile, with the Fubini-Lipatov scale optimized by golden-section search.
- *Phase diagrams.* Constructed stability phase diagrams in the (M_h, M_t) plane from **million-point OpenMP-parallel parameter sweeps**, characterizing where the numerical treatment deviates from the analytical approximation near the metastability boundary.
- *Diagnosis.* Diagnosed the **unphysical negative bounce actions** of the strict Fubini-Lipatov ansatz, tracing them to the inconsistency between its constant-quartic kinetic term and the running-coupling potential.
- *Cosmological hierarchy problem.* Modeled the Higgs field as a **stochastic spectator** during slow-roll inflation, studying whether **volume-weighted anthropic selection** can bias Higgs initial conditions toward the metastable electroweak vacuum without introducing new low-energy degrees of freedom.

Tests of General Relativity in Pseudo-Newtonian Frameworks — Jan 2024 – May 2026

*Supervisor: Prof. Banibrata Mukhopadhyay, Indian Institute of Science, Bengaluru | Published in Eur. Phys. J. C **86**, 1280 (2026)*

- *Analysis.* Derived analytical expressions for **perihelion precession**, **gravitational redshift** and **gravitational lensing** in several pseudo-Newtonian potentials, including the Paczyński-Wiita, Artemova-Björnsson-Novikov and Mukhopadhyay potentials, and compared their predictions with Schwarzschild and Kerr general relativity.
- *Computation.* Developed a **Python framework** for the numerical analysis and visualization of orbital observables, generating the figures used to benchmark multiple pseudo-Newtonian models across parameter regimes.
- *Result.* Identified the regimes in which parameterized pseudo-Newtonian potentials successfully reproduce individual relativistic effects, and showed that **no single potential consistently captures** all three simultaneously.

Independent Reading: Lorentz Representations and the Little-Group Formalism — May – Jun 2025

Self-directed reading project, supervised by Prof. Rick Sandeepan Gupta, TIFR, Mumbai

- *Foundations.* Studied foundational quantum field theory, focusing on Lorentz and Poincaré group representations and the **little-group formalism** for massive and massless particles.
- *Continuous spin.* Investigated **continuous-spin representations**, studying the formalism developed by Schuster and Toro and analyzing its implications for exotic particle states beyond the standard discrete spins.

Experimental Projects

Traveling Flame in Narrow Tubes — 2024

International Physicists' Tournament (IPT), IISc team, Bengaluru

- *Apparatus.* Designed and **3D-printed narrow tubes** with controlled obstacles to study flame propagation.
- *Measurements.* Conducted experiments characterizing ignition thresholds, flame-speed regimes and the transition from **laminar deflagration to turbulent combustion**.
- *Diagnostics.* Analyzed high-speed footage and pressure data to quantify turbulence onset, providing insight into premixed-flame behavior in confined geometries.

Jumping Hoop Phenomenon — 2023

International Physicists' Tournament (IPT), IISc team, Bengaluru

- *Model.* Developed a theoretical model for the vertical jumping of a rigid hoop driven by an off-center sliding puck, incorporating rotational inertia, impulsive normal forces and the shifting center of mass.
- *Measurement.* Performed controlled experiments measuring jump heights and trajectories under varying mass ratios and initial conditions.
- *Simulation.* Implemented **numerical simulations in C++** to solve the coupled equations of motion, achieving good agreement with the experimental observations.

Honors and Awards

PLANCKS 2024	Represented India at the IAPS PLANCKS international finals in Dublin, Ireland, placing **11th worldwide**; re-selected for the 2025 competition.
JEE 2023	Joint Entrance Examination: **All India Rank 33** in JEE Advanced (~180,000 candidates) and AIR 281 in JEE Main (~1.1 million candidates).
KVPY Fellow	**All India Rank 171** (SA stream) in the Kishore Vaigyanik Protsahan Yojana, a national scholarship for research careers in the basic sciences.
NIUS Physics Scholar	Batch 21.1: selected by HBCSE-TIFR to a national program admitting approximately 70 undergraduates annually for advanced research training.
IPT 2023, 2024	International Physicists' Tournament, Indian national round: **national runner-up** with the IISc team in both years; contributed theoretical modeling and experimental design.
INAO 2021, 2022	Indian National Astronomy Olympiad: **selected twice for the Orientation-cum-Selection Camp (OCSC)**, the final national stage of selection for India's team at the International Olympiad on Astronomy and Astrophysics.
RMO and NSEP	Top 1% state ranking in the Regional Mathematical Olympiad (2020, 2021) and the National Standard Examination in Physics (2020).

Summer Schools and Programs

Perimeter Institute Undergraduate Summer School (USS) — Jun 2026

Perimeter Institute for Theoretical Physics, Waterloo, Canada

NIUS Physics Camp — Jun 2024

Homi Bhabha Centre for Science Education (HBCSE-TIFR), Mumbai

Relevant Coursework

Advanced theoretical physics	**Quantum Field Theory I & II**, **Standard Model of Particle Physics**, **General Relativity**, Advanced Mathematical Physics, Nuclear and Particle Physics
Core physics & astrophysics	Quantum Mechanics I & II, Classical Mechanics, Statistical Mechanics, Electrodynamics, Condensed Matter Physics, Thermal Physics, Fundamentals of Astrophysics
Mathematics	Linear Algebra and Analysis, Probability and Statistics, Mathematical Methods of Physics

Technical Skills

Programming	C++ (including OpenMP parallelization), Python
Scientific computing	Multi-loop RGE evolution, adaptive ODE integration, numerical optimization, large parameter scans, scientific visualization
Libraries and tools	NumPy, SciPy, Matplotlib, SymPy, Mathematica; Git, LaTeX